

Months	Weeks	MDM	Modulators	PD and System Level
September	1		Kickoff & initial literature search (survey papers).	
October	2		- Understanding of performance metrics (V_{π}, BW, IL, ER, FWHM, Q Factor). - Simulate basic Mach Zender Modulator (MZM) optical response (transfer function). - Simulate basic Microring Modulator (MRM) optical response (transfer function). - Compare MZM vs MRM.	
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November	6		- summary of performance metrics (V_{π}, BW, IL, ER, FWHM, Q Factor). - Simulate basic MRM optical response at different biasing points. - Deep literature review (10 core papers) in both MRM and MZM.	
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December	10		- literature review summary. - Shortlist candidate modulators (ring modulators) and trade-off table. - decide final device to pursue.	
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January				
February	1		- Start device-level simulation for our chosen modulator variant. - Finalize modulator design and geometry.	
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March	5		- Finalize electrical/optical co-simulation (EO response). - Decide the proper operation BW. - Produce performance metrics plots.	
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April	9		- performance review. - Analyze measurement data - Generate final design. - Start writing Methods section for documentation (simulation tools, parameters, settings).	
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May	13		Finalize full documentation: Introduction + Methods + Results + Discussion + Conclusion.	
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